

Lesson 22 Inverse Function ; Function Composition

Lesson 21: $f: A \rightarrow B$, the inverse f^{-1} is a fn from B to A iff f is a bijection

Def A fn $f: A \rightarrow B$ is invertible if the inverse relation f^{-1} is a fn from B to A , and f^{-1} is the inverse (function) of f .

Ex Last time: $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x$ is bijective

f is invertible

$$f^{-1}: \mathbb{R} \rightarrow \mathbb{R}, \quad f^{-1}(x) = ?$$

$$\left\{ \begin{array}{l} x \xrightarrow{f} 2x \\ x \xleftarrow{f^{-1}} 2x \\ \frac{x}{2} \xleftarrow{f^{-1}} x \end{array} \right.$$

$$f = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y = 2x\}$$

$$f^{-1} = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x = 2y\}$$

$$\{(y, x) \in \mathbb{R} \times \mathbb{R} : y = 2x\}$$

$$= \{(x, y) \in \mathbb{R} \times \mathbb{R} : y = \frac{x}{2}\}$$

Onto: let $y \in \mathbb{R}$

$$\begin{aligned} f(x) &= y \\ 2x &= y \rightarrow x = \frac{y}{2} \in \mathbb{R} \end{aligned}$$

$$= \widetilde{f^{-1}(y)}$$

$$\boxed{\begin{array}{l} y = f(x) \\ x = f^{-1}(y) \end{array}}$$

$$f^{-1}(y) = \frac{y}{2} \xrightarrow{x}$$

$\downarrow$
 x

$$\boxed{f^{-1}(x) = \frac{x}{2}}$$

Prop If f is invertible, then f^{-1} is invertible and $(f^{-1})^{-1} = f$.

Proof: Notes

Function composition

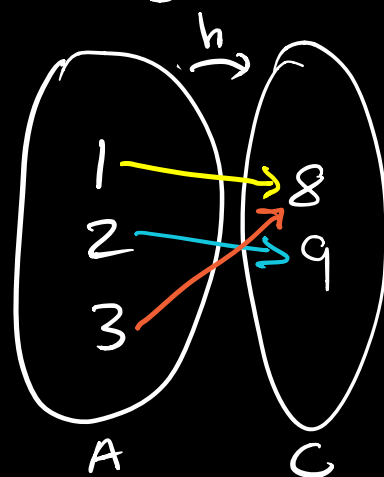
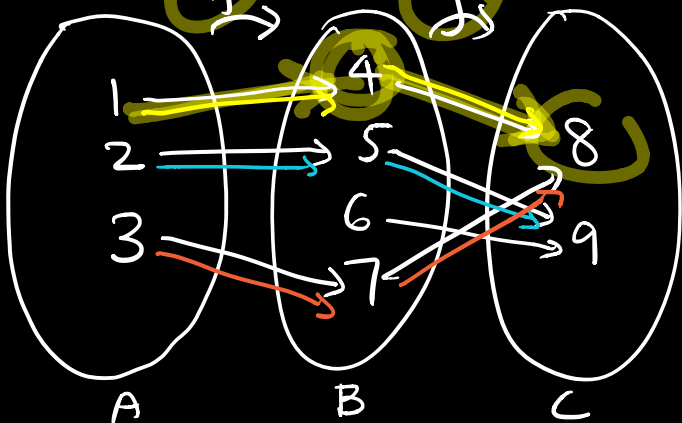
$$f: \underline{A} \rightarrow \underline{B}, g: \underline{B} \rightarrow \underline{C}$$

The composition of f and g is the fn:

$$g \circ f: \underline{A} \rightarrow \underline{C}, \quad (g \circ f)(x) = g(f(x))$$

Ex $f: A \rightarrow B, g: B \rightarrow C$

$$h = g \circ f: A \rightarrow C$$



$$\begin{aligned} h(1) &= (g \circ f)(1) \\ &= g(f(1)) \\ &= g(4) \\ &= 8 \end{aligned}$$

$$h(2) \dots$$

$$h(3) \dots$$

Ex $f: \mathbb{R} \rightarrow \mathbb{R} \quad f(x) = x^2 + 1$

$$g: \mathbb{R} \rightarrow \mathbb{R} \quad g(x) = 2x - 3$$

$$\begin{aligned} g \circ f: \mathbb{R} \rightarrow \mathbb{R}, \quad (g \circ f)(x) &= g(f(x)) \\ &= g(x^2 + 1) \\ &= 2(x^2 + 1) - 3 \end{aligned}$$

2nd fn
1st fn

$$\begin{aligned}
 & \downarrow \quad \downarrow \\
 & f \circ g : \mathbb{R} \rightarrow \mathbb{R}, \\
 & (f \circ g)(x) = f(g(x)) = f(2x-3) \\
 & \quad \quad \quad = (2x-3)^2 + 1
 \end{aligned}$$

$\neq$

fn composition is not commutative:

$$f \circ g \neq g \circ f \quad (1+2)+3 = 1+(2+3)$$

fn composition is associative:

$$(h \circ g) \circ f = h \circ (g \circ f)$$

Proof same domain, codomain, fn rule

We want to show that $[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x)$

$$[(h \circ g) \circ f](x) = (h \circ g)(f(x))$$

$$= h(g(f(x)))$$

$$[h \circ (g \circ f)](x) = h((g \circ f)(x))$$

$$= h(g(f(x)))$$

If f and g are onto, then g ∘ f is onto

$$f : A \rightarrow B, \quad g : B \rightarrow C$$

$$g \circ f : A \rightarrow C, \quad (g \circ f)(x) = g(f(x))$$

Let $z \in C$, want to show there is
 $x \in A$ s.t. $(g \circ f)(x) = z$.

If $g \circ f$ is onto, then g is onto.

Contrapositive: If g is not onto, then
 $g \circ f$ is not onto.

If g is not onto, there is $z \in C$ s.t.
 $g(y) \neq z$ for any $y \in B$.

Want to show that $(g \circ f)(x) \neq z$
for any $x \in A$